



Orchestration, Aggregation and Monitoring of Federated Workflows

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Summary: BioChef runs analytic workflows through agents distributed across institutions, but has no central system to coordinate them. This work builds the BioChef Coordination Hub: a researcher submits an analysis once, the Hub distributes it across federated sites, aggregates the results under differential privacy with noise-budget management, and monitors the whole infrastructure in real time. It also provides a library of reusable workflows, validated across at least three simulated sites.

Objectives / Work done / Results

Objectives

- Orchestrate workflows across agents at each site handling queues, retries, dependencies
- Aggregate results with differential privacy and noise.
- Real-time monitoring like progress, alerts, searchable logs
- Validate on 3+ simulated sites.

Work done

- Studied the BioChef Front-End and Agent repositories
- Understand BioChef's agent architecture and workflow model

Results

- Good understanding of the project: what needs to be built, why it needs to be built and how the other components work

Next Steps / Challenges / Bottlenecks

Next Steps

- Specify the Hub–agent API

Challenges

- Securing agent–Hub communication across institutions

Bottlenecks

- Limited access to real multi-site test setups
- Coordination with the existing IEETA codebase